A Sidelobe-Aware Small Ship Detection Network for Synthetic Aperture Radar Imagery

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***Abstract*— Ship detection from synthetic aperture radar (SAR) remote sensing images is essential for monitoring water traffic and marine safety. Numerous methods for ship detection have been developed; however, the detection of small ships presents unique challenges. SAR image characteristics, such as the sidelobe effect and blurred outline induced by the special imaging mechanism, as well as the small ship size, are the primary factors that lower the detection accuracy. This article provides a sidelobe-aware small ship detection network for SAR imagery. First, considering the sidelobe effect and blurred outline, dual-pooling, i.e., average pooling and max pooling, was utilized to build a feature extraction module that lowered the effects of strong scattering points outside of the ship body and enhanced the ship body information. Second, as the bipartition process of the average pooling and maximum pooling caused some loss of original data information, different feature maps in the network were concatenated to construct a new network structure to compensate for the information lost and enrich the small ship features. Third, because the traditional loss function based on centroid distance and aspect ratio may result in the same loss function value for different prediction box sizes, a novel loss function based on the dual Euclidean distances of the corner point coordinates between the prediction box and the real box was proposed, which could accurately describe various overlapping box situations. Experiments using the Large-Scale SAR Ship Detection dataset (LS-SSDD), the SAR Ship Detection dataset (SSDD), and the AIR-SARShip dataset validated the efficacy and state-of-the-art performance.**

***Index Terms*— Feature extraction, regression calculation of bounding box, ship detection, sidelobe, synthetic aperture radar (SAR).**

I. Introduction

THE detection of ships from remote sensing images has drawn a great deal of attention with the rapid advancement of remote sensing technology, which is crucial for monitoring water traffic and ports in coastal areas and ensuring marine safety. Synthetic aperture radar (SAR) is one of the most essential remote sensors that can collect target images in almost any weather and at any time of day. It is widely used in various application areas, such as target detection and recognition [1], [2], [3], [4], quantitative parameter inversion [5], and ground surface segmentation and classification [6], [7], [8].

Numerous ship detection methods have been proposed, which can be roughly categorized as traditional or deep learning-based detection methods [9]. The constant false alarm rate (CFAR) is one of the most widely utilized traditional detection methods [10], [11]. Based on the statistical distribution of clutter, a threshold is calculated while maintaining a certain false alarm rate. However, due to the interference of background clutter, it is difficult to guarantee a very low false alarm rate with high recall [12]. Migliaccio et al. [13] make use of a new physical SAR processing methodology. By constructing a unitary physical model and combining it with CFAR, the problem of the effect of clutter on target detection is improved.

With the advancement of deep learning technology, numerous deep learning-based detection methods and datasets have been developed [14], [15], [16], [17]. The convolutional neural network (CNN) is one of the most commonly used networks for deep learning. Various CNN-based detection methods have been developed and achieved better results in various scenarios. CNN-based detection methods offer the advantage of being able to handle more complex scenarios and achieve higher detection accuracy. Representative two-stage detection methods include Region-CNN (R-CNN), Fast R-CNN [18], cascaded R-CNN [19], and so on. The advantage of the two-stage detection method is that the predicted position of the bounding box is accurate. The Single-Shot MultiBox Detector (SSD) [20] and You Only Look Once (YOLO) [21] series, as representatives of one-stage detection methods, also perform well. Since there is no coarse extraction in the one-stage detection methods, their detection speeds are increased. Some researchers have conducted deep learning-based research in the field of remote sensing, which employs currently advanced deep learning models or artificial intelligence (AI) models in remote sensing. Sun et al. [22] propose a Remote sensing foundation Model framework called RingMo, which enhances the processing ability of the foundation model for remote sensing data. He et al. [23] propose a Transformer-Induced Hierarchical Graph Network (GraFNet) for multimodal semantic segmentation in remote sensing scenes. He et al. [24] propose an end-to-end online framework, which is called

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TABLE I
SUMMARY OF REPRESENTATIVE SOLUTIONS FOR SMALL SHIP DETECTION IN RECENT YEARS

Method	Year	Improved Feature Extraction	Multi-Scale Fusion	New Network Structure	Improved Anchor
BiFA-YOLO [27]	2021	-	-	●	●
DB-YOLO [28]	2021	-	●	●	-
N-YOLO [29]	2021	●	-	●	●
FASCNet [30]	2022	●	-	●	-
R-CenterNet+ [31]	2022	●	-	●	●
Quad-FPN [32]	2022	●	-	●	-
Lite-YOLOv5 [33]	2022	●	-	●	-
S-LPAN [34]	2022	●	●	-	-
FSFADet [35]	2022	●	-	●	●
BANet [36]	2022	●	●	-	●
MSIF [37]	2022	●	●	-	-
SAR-Net [38]	2022	-	-	●	-
Ours	2023	●	-	●	●

Tracking via Graph-based Multitask Reasoning (TGraM), for multiobject tracking in satellite videos. Zhou et al. [25] present an improved SSD model for SAR target detection. Chang et al. [26] offer an enhanced GPU-based YOLOv2 for ship detection in SAR images. In general, current deep learning-based detection methods have achieved satisfactory accuracy for medium and large ships.

In this article, small ship detection is considered from a deep learning perspective. Detecting small ships from SAR images still presents significant challenges due to the SAR special imaging mechanism, the small ship size, and the weak scattering intensity. Those challenges result in two main algorithmic issues. The first one relies on the fact that common detection layers with large receptive fields are unsuitable for detecting small ships. The second one depends on the effects of clutter and sidelobes that make the detection of smaller ships difficult and may generate missed targets. This makes the network even worse at capturing information about small ships and reduces the performance of the target detection results. To address these challenges, researchers have proposed various enhancements, including improved feature extraction, multiscale learning, new network structures, and improved anchors, as summarized in Table I.

1) *Improved Feature Extraction [29], [30], [31], [32], [33], [34], [35], [36], [37]:* Yu et al. [30] propose a fast and lightweight detection network (FASC-Net) that shortens the information transmission path, enhances the precise location information contained in the underlying feature map, and enables the feature map to express more adequate information in the network. Jiang et al. [31] propose the lightweight anchor-less detector R-Centernet+ and use CBAM-DLA34 for fine extraction of ship features. Zhu et al. [35] propose an orientation-independent ship detection method (FSFADet) based on the separation alignment of features. Zhang et al. [37] offer a multisize inference fusion (MSIF) framework that integrates two false alarm elimination methods, pixel feature filtering and multi-source fusion, to eliminate false alarms and improve the overall performance of ship detection in large-scale SAR images. Zhang et al. [32] suggest quadratic features of feature pyramid networks (FPNs) for SAR ship detection, which addresses the issues of complicated background interference, considerable variations in ship features, and blurred features of small ships.
2) *New Network Structure [27], [28], [29], [30], [31], [32], [33], [35], [38]:* Tang et al. [29] design N-YOLO based on the YOLO algorithm and combined it with a noise classifier to improve its capability of image feature extraction. Sun et al. [27] develop a novel YOLO-based arbitrary-oriented SAR ship detector that employs bidirectional feature fusion and angular classification (BiFA-YOLO) to solve the problem of multiscale, arbitrary orientation, and dense arrangement of prominent features. Xu et al. [33] provide a lightweight Lite-YOLOv5 that ensures detection performance while reducing the model size and floating-point operations. Gao et al. [38] redesign the CNN architecture based on YOLOv4 with a specific skill fusion strategy and propose an SAR-Net detection network to enhance the performance of small ship detection.
3) *Multiscale Learning [28], [34], [36], [37]:* Zhu et al. [28] develop an SAR ship detector called repetitive bilateral YOLO (DB-YOLO), which redesigns the parameter distribution in the network and improves the multiscale detection capabilities of ships through feature multiplexing. Zhou et al. [34] propose the S-LPAN detection approach, which makes use of dual-polarimetric SAR data and polarization decomposition features.
4) *Improved Anchor [27], [29], [31], [35], [36]:* Hu et al. [36] suggest an anchorless strategy to replace the anchor-based approach commonly used in SAR ship detection algorithms and propose a BANet detection network, based on which a multiscale ship detection method and deformable convolution are introduced to capture the local information of ship targets efficiently.

Although numerous attempts have been made to improve the detection performance of small ships in SAR images, the majority of these efforts are based solely on the features of SAR images and focus on improving only the generic visual detection models, rather than taking into account the unique SAR imaging mechanisms [39]. SAR actively transmits

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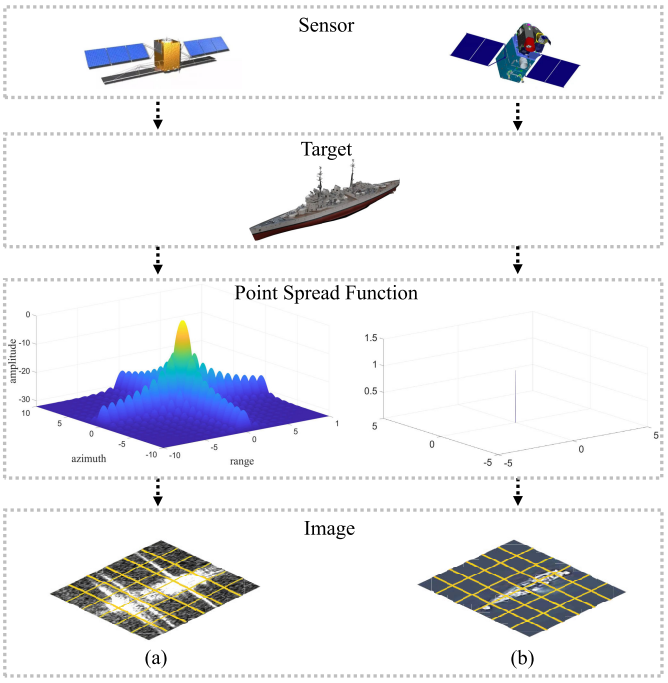


Fig. 1. (a) SAR image and (b) optical image of a ship.

microwave signals, receives microwave signals reflected from the target, and generates focused images using the signal processing method of matched filtering, which inevitably leads to strong scattering points and sidelobes in SAR images [40]. The imaging process of SAR differs greatly from that of optical sensors, resulting in a target image with unique characteristics. Fig. 1 depicts an optical and SAR image of a ship. In optical satellite image imaging, for a pixel point that we can liken to an impulse response, the ship's outline can be clearly shown. Its SAR image contains numerous strong scattering points, forming a strong sidelobe effect, which will have a greater impact on the ship detection accuracy than the optical image. The influence of the sidelobe will be more significant due to the small size and weak scattering of small ships, as well as the interference from background clutter, such as ground and coastal waves. Therefore, it is not sufficient to directly transpose the deep learning target detection algorithm used on optical images to SAR images. We need to solve the problems on SAR images from the perspective of physical formation to improve the detection performance.

Numerous researchers have examined the role of SAR imaging mechanisms in various SAR applications. Guo et al. [41] consider the strong scattering points of aircraft and propose a new hybrid approach of scattering information enhancement and an attention pyramid network (APN) for aircraft detection. Kang et al. [42] propose an anchor-free scattering feature relation (SFR) detection network for the structural and spatial relationships of important scattering points, which integrates the analysis of target scattering point distribution and the background of environmental information to enhance the target saliency under different imaging conditions. Sun et al. [43] propose a novel method named strong scattering points network for ship detection, which extracts the strong scattering points in SAR images, captures the multiscale information of ships to improve the location accuracy, and suppresses the interference of strong scattering points on land. Wang et al. [44] investigate the impact of sidelobes on SAR target detection and propose a method based on a fast block constant false alarm detector. However, the feature analysis method does not apply to complex scenes. Xu et al. [45] analyze the sidelobe suppression method, propose additional insights into scattering noise, and optimize a weighting model to reduce the SAR sidelobe using a bionic method with the same imaging resolution.

Although a lot of efforts have been made to investigate the role of SAR imaging mechanisms, most of them focus on how to suppress SAR scattering features, such as sidelobes, or how to consider the sidelobe effect in traditional detection methods. The impact of SAR imaging mechanisms on deep learning-based small ship detection applications is still a topic worth investigating.

In this article, the role of SAR imaging mechanisms in small ship detection is investigated. An improved feature extraction approach considers the unique 2-D-sidelobe effects, a corresponding new network structure, and an improved anchor are designed. Specifically, due to the SAR imaging process of matched filtering, the ship information is distributed across the ship hull to form sidelobes, which will affect the detection performance. Considering that average pooling takes the mean value from a region's pixels and weakens the higher value in the region, while maximum pooling extracts the maximum value from a region's pixels, this article designs a feature extraction approach taking into account the horizontal and vertical pixels, and reduces the impact of scattering points on the outer side of the ship by average pooling while enhancing the extraction of the ship's main body information with maximum pooling. Second, some of the feature information from the original image is lost because the proposed feature extraction approach uses average pooling and maximum pooling. To solve this problem, a new network structure is designed inspired by the bidirectional feature fusion (BiFF) [27], which transfers the lower feature layer information to the deeper feature layers to achieve bottom-up fusion of features. In the new network structure, before feature extraction, the feature map is concatenated with the feature map of the neck network to extract the background information from the original image and concatenate the shallow and high-level features to enhance the ship's feature information. Third, the current regression loss function for the bounding box does not fully account for the various situations. Bounding boxes with the same centroid distance and aspect ratio may also yield different loss values, which affects the ship detection performance. In addition, the judgment of the centroid of the bounding box will be affected by sidelobe interference. To solve these problems, a novel loss function based on the dual Euclidean distances of the corner point coordinates between the prediction box and the real box is proposed, which makes up for the deficiencies of the existing loss function while optimizing the ship detection model.

In brief, the primary contribution points of this article are given in the following.

1) In order to consider the influence of the sidelobe, a feature extraction with a sidelobe awareness module using

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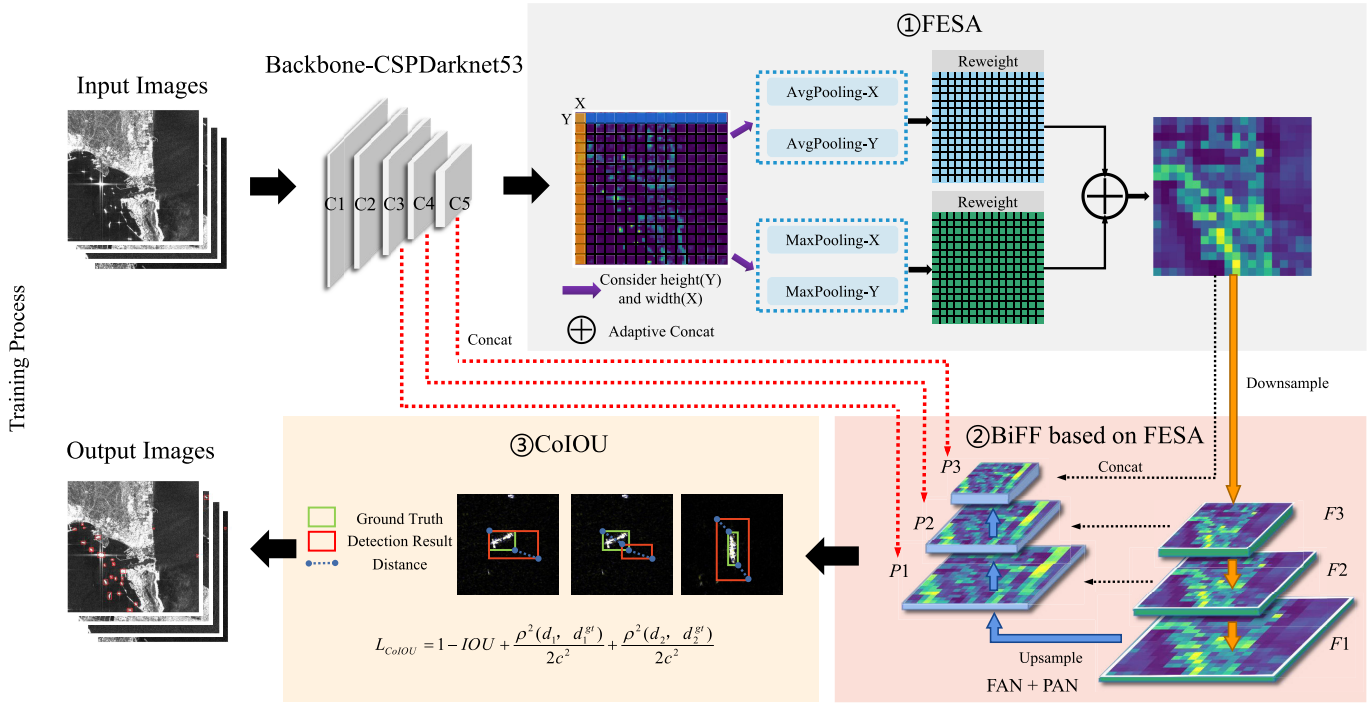


Fig. 2. Overall structure of the proposed network. (① FESA; ② BiFF Based on FESA; and ③ CoIOU loss function.)

dual pooling (average pooling and maximum pooling) is presented. It can reduce the influence of sidelobes and increase the detection model's ability to gather information about the ship's main body.

2) In order to solve the problem of information loss induced by average pooling and maximum pooling and fully extract small ship features, the feature maps generated by the backbone network are concatenated to the neck network, which is based on the structure of the FPN and pixel aggregation network (PAN). It can compensate for the information lost during dual-pooling and improve small ship features.
3) In order to overcome the defects in the current loss function of anchors, this article proposes a new loss function calculation method by using the dual Euclidean distances between corner points of the bounding boxes instead of the centroids. It can reduce the sidelobe influence, ship misjudgment, and false alarms.

The remainder of this article is structured as follows. Section II presents the proposed method. Section III validates the proposed method by comparing it with other methods. Section IV draws conclusions.

II. Method

Fig. 2 shows the overall structure of the proposed model. After the backbone module, a feature extraction with sidelobe awareness (FESA) module guided by the 2-D-impulse response function (IRF) is added. Then, the FPN [46] and PAN [47] network structures are retained, and a BiFF network structure based on FESA is added. In addition, a new bounding box regression loss function CoIOU is used.

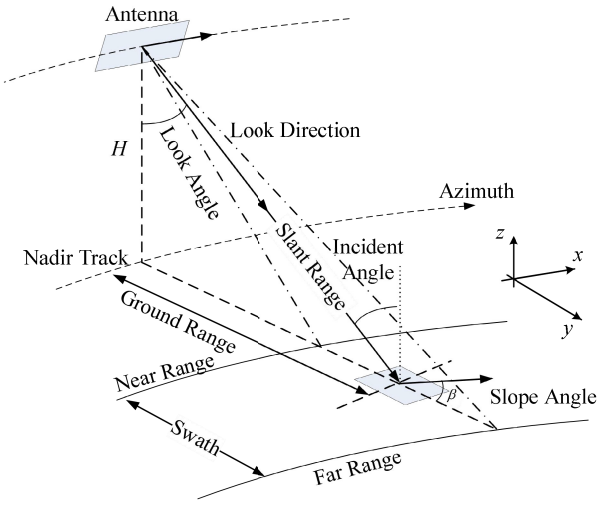


Fig. 3. SAR imaging geometry.

A. Feature Extraction With Sidelobe Awareness

1) SAR Imaging Mechanism: SAR is an active imaging sensor that emits microwave signals and then receives the backscattering echo signals from the targets. Fig. 3 illustrates the SAR imaging geometry. The signal emitted by SAR can be expressed as

$$\boldsymbol{E}_t(t) = \hat{\boldsymbol{p}}_t g_{\mathrm{rg}}(t) \mathrm{e}^{j\omega_0 t} \tag{1}$$

where $\hat{\boldsymbol{p}}_t$ represents the polarization of the transmitting antenna. $g_{\mathrm{rg}}(t)$ represents the ideal linear frequency-modulated (LFM) pulse signal. t represents the cross-track time. ω_0 represents the center frequency. After coherent demodulation of the

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SAR received signal, the final received signal can be expressed as

$$u_{rj}(x,t) = A_j \hat{\boldsymbol{p}}_r^{\mathrm{T}} \boldsymbol{F}_j \hat{\boldsymbol{p}}_t g_{\mathrm{rg}}\left(t - \Delta t_j\right) \mathrm{e}^{-j\omega_0 \Delta t_j} \tag{2}$$

where A_j represents the decay of the backscattering amplitude matrix $\boldsymbol{F}_j$, $\boldsymbol{F}_j$ represents the scattering amplitude matrix of j, and $\hat{\boldsymbol{p}}_r$ represents the polarization of the receiving antenna. Δt_j represents the round-trip time delay, and r_1 and r_j represent the positions of the antenna and the target, respectively.

The general model of the signal received from the ground is equivalent to the 2-D convolution of the ground reflectivity and the SAR system impulse response. The system impulse response of a point target is

$$h(x,t) = \left[\mathrm{e}^{-j\omega_0 \Delta t_j} \delta\left(t - \Delta t_j\right)\right] \otimes \left[g_{\mathrm{rg}}(t)\right] \tag{3}$$

where δ represents the impulse function and $\otimes$ represents the convolution.

Provided that the approximation conditions are satisfied, both $\mathrm{e}^{-j\omega_0 \Delta t_j} \delta\left(t - \Delta t_j\right)$ and $g_{\mathrm{rg}}(t)$ can be dimensionalized by approximating them as linear FM functions that are not coupled to each other. SAR imaging processing focuses on the received echo data onto a point target. Since the ideal point target response (impulse function) has an infinite bandwidth, while the bandwidth of a real radar system is always limited, the IRF of an SAR system is defined as the 2-D response to a point target while ignoring background clutter and thermal noise. In fact, the focused point target recompresses the impulse response, compressing (4) into a 2-D sinc function

$$\mathrm{E}(x,t) = \begin{cases} \left[\dfrac{\sin\left(\pi \frac{t}{\tau_r}\right)}{\pi \frac{t}{\tau_r}}\right] \cdot \left[\dfrac{\sin\left(\pi \frac{x}{\rho_\alpha}\right)}{\pi \frac{x}{\rho_\alpha}}\right], & t \neq 0, x \neq 0 \\ 1, & t = 0, x = 0 \end{cases} \tag{4}$$

where τ_r represents the mainlobe width of azimuthal direction, t represents the time changing, x represents aerial carrier location, and ρ_α represents the mainlobe width of range direction.

In SAR images, resolution refers to the minimum distance at which two targets can be distinguished. Using the aforementioned sinc function as an example, the absolute value is taken and converted into decibels when analyzing the SAR imaging performance index, and the waveform is shown in Fig. 4. The sidelobes formed in the imaging system contain more clutter information, and the cross interference formed by the sidelobes has a greater impact on the information extraction from the ship and can affect the performance of ship detection.

2) Feature Extraction With Sidelobe Awareness: Due to the SAR special imaging mechanism, the generated high-resolution SAR ship images will be affected by the strong sidelobes. The ship's features will be dispersed across a few pixel points around the ship. It is challenging to accurately extract the feature relationships in the channels by using only the convolution operation. As seen from the convolutional block attention module (CBAM) [48] and coordinate attention [49], average pooling and maximum pooling can be effective in extracting feature information. The effective

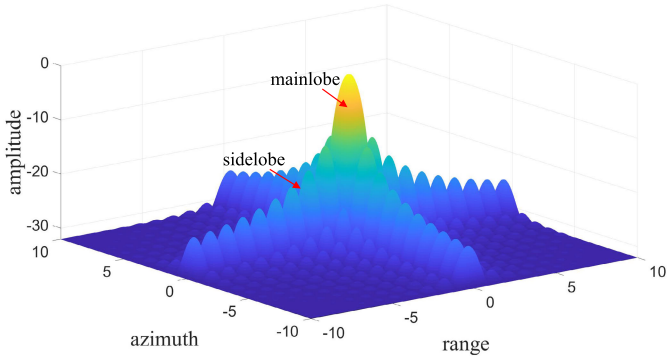


Fig. 4. Illustration of the 2-D-IRF.

extraction of local information from the ship image can be performed from different directions.

In this article, a feature extraction module considering 2-D-IRF characteristics is proposed. As shown in Fig. 5, the input feature maps are handled in two directions, i.e., height and width directions. The yellow dots indicate the ship's primary information, while the light blue dots represent the scattered information due to sidelobes and other interference. The green dots represent the information reweighted in the height direction by average pooling and maximum pooling, while the red dots represent the information reweighted in the width direction by average pooling and maximum pooling. The purple dots represent the feature pixel information obtained by FESA. Pooling from both the width and height directions can enhance the extraction of local information on each channel. The dual-pooling strategy proposed in this article can assist the model in reducing the effect of sidelobes and extracting more feature information.

This article refers to the theoretical advantages of squeeze-and-excitation (SE) networks [50] for global information embedding and channel relationship adaptive recalibration. Given the input feature map, each channel is chosen to extract features in the X- and Y-directions, i.e., in the width and height directions, respectively, and both average pooling and maximum pooling are considered for the 1-D feature encoding operation. Average pooling can enhance the main feature information of the ship body to a certain extent and weaken the effect brought by the sidelobes. Maximum pooling can extract the overall information of the ship hull. Specifically, given the input x, two spatial segments $(H, 1)$ or $(1, W)$ of the pooling kernel are used in this article to encode each channel along with the horizontal and vertical coordinates, respectively. Thus, by the dual-pooling, the output in channel c can be expressed as

$$z_{\mathrm{Avg_}h}(h) = \mathrm{Conv1D}_{\mathrm{Avg}}(x_c(h,i)) \tag{5}$$

$$z_{\mathrm{Avg_}w}(w) = \mathrm{Conv1D}_{\mathrm{Avg}}(x_c(j,w)) \tag{6}$$

$$z_{\mathrm{Max_}h}(h) = \mathrm{Conv1D}_{\mathrm{Max}}(x_c(h,i)) \tag{7}$$

$$z_{\mathrm{Max_}w}(w) = \mathrm{Conv1D}_{\mathrm{Max}}(x_c(j,w)) \tag{8}$$

where $\mathrm{Conv1D}_{\mathrm{Avg}}$ represents the average pooling operation in one dimension and $\mathrm{Conv1D}_{\mathrm{Max}}$ represents the maximum pooling operation in one dimension.

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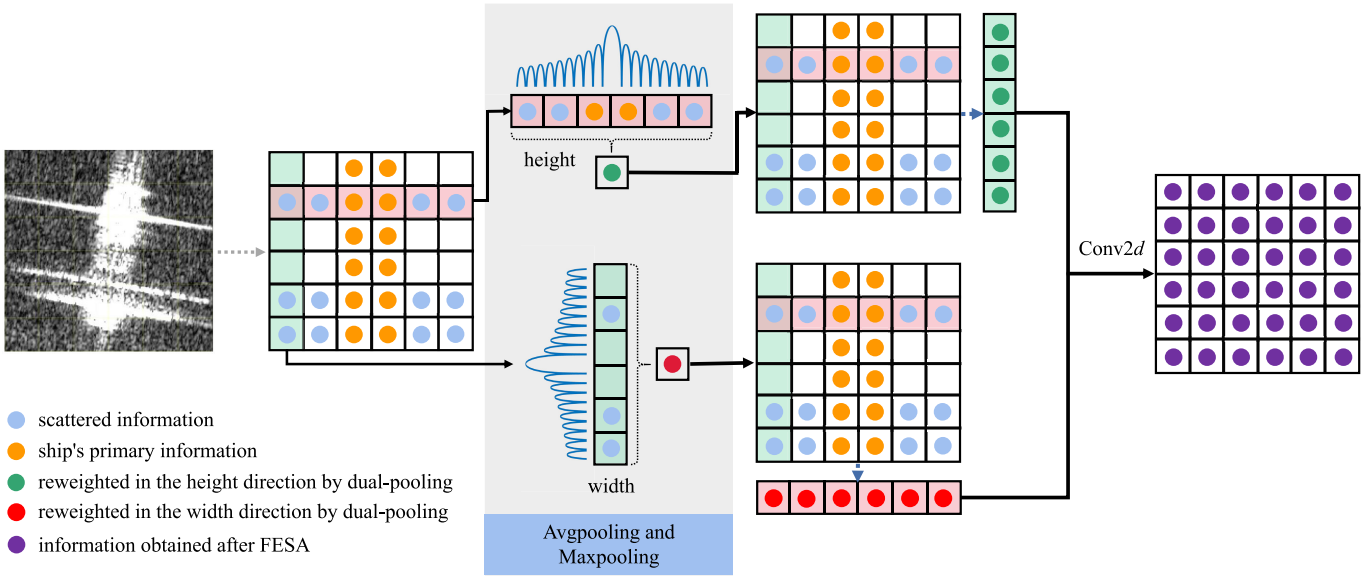


Fig. 5. Overall structure of the proposed FESA.

For different pooling methods, the features are aggregated to two different spatial directions of the feature maps, and a set of perceptual feature maps in the width and height directions is output. The corresponding feature maps become $z_{\text{Avg_}h} \in R^{C*H*1}$, $z_{\text{Avg_}w} \in R^{C*1*W}$, $z_{\text{Max_}h} \in R^{C*H*1}$, and $z_{\text{Max_}w} \in R^{C*1*W}$ after pooling, where C, H, and W denote the number of channels of the input feature map, the height of each channel feature map, and the width of each channel feature map, respectively.

After the pooling operation, the feature information expressed in the global receiver domain of the SAR image is obtained. To make better use of these features, simple concatenation by dimension and convolution operations are used in this article. This not only makes full use of the captured position information to accurately highlight the region of interest but also effectively captures the relationship between individual channels. Using the obtained $z_{\text{Avg_}h}$, $z_{\text{Avg_}w}$, $z_{\text{Max_}h}$, and $z_{\text{Max_}w}$, this article lets the features be obtained by the same pooling be stitched together, respectively, and then by a convolutional transformation function

$$f_{\text{Avg}} = \delta(F_1([z_{\text{Avg_}h}, z_{\text{Avg_}w}]) \tag{9}$$

$$f_{\text{Max}} = \delta(F_1([z_{\text{Max_}h}, z_{\text{Max_}w}]) \tag{10}$$

where $F_1([\cdot,\cdot])$ is denoted as the splicing operation along the spatial dimension followed by a transformed convolution function, δ is for the nonlinear activation function, and the obtained $f \in R^{C/r*(H+W)}$ is the intermediate feature mapping that encodes the spatial information of the width and height of the feature map. Here, r is the reduction ratio for controlling the block size as in the SE block. By controlling the block size reduction ratio in the module, the intermediate feature mappings obtained by average pooling and maximum pooling are then decomposed into two separate tensors; $f^h_{\text{Avg}} \in R^{(C/r*H)}$, $f^w_{\text{Avg}} \in R^{(C/r*W)}$, $f^h_{\text{Max}} \in R^{(C/r*H)}$, and $f^w_{\text{Max}} \in R^{(C/r*W)}$. Then, the other two 1×1 convolution transforms are used to transform f^h_{Avg}, f^w_{Avg}, f^h_{Max}, and f^w_{Max} into tensors with the same number of channels to the input X, respectively,

$$\mathbf{g}^h_{\text{Avg}} = \sigma\left(F\left(f^h_{\text{Avg}}\right)\right) \tag{11}$$

$$\mathbf{g}^w_{\text{Avg}} = \sigma\left(F\left(f^w_{\text{Avg}}\right)\right) \tag{12}$$

$$\mathbf{g}^h_{\text{Max}} = \sigma\left(F\left(f^h_{\text{Max}}\right)\right) \tag{13}$$

$$\mathbf{g}^w_{\text{Max}} = \sigma\left(F\left(f^w_{\text{Max}}\right)\right) \tag{14}$$

where σ is the sigmoid activation function. The output parameters are then expanded and used as attention weights. Then, for the results obtained from average pooling and maximum pooling, this article splices them after feeding them through a convolution function and reweighting sizes. The final output is

$$F = \text{Conv2D}\left(F_1\left(\left[\mathbf{g}^h_{\text{Avg}}, \mathbf{g}^w_{\text{Avg}}\right]\right), F_2\left(\left[\mathbf{g}^h_{\text{Max}}, \mathbf{g}^w_{\text{Max}}\right]\right)\right) \tag{15}$$

where Conv2D represents the 2-D convolution operation.

B. Bidirectional Feature Fusion Based on FESA

In high-resolution SAR images, the ship size varies greatly, and the existing feature fusion network can hardly meet the needs of multiscale ship feature extraction. In addition, although dual-pooling can weaken the sidelobe effect to some extent, it causes some information loss. For ships with small sizes, the information loss will have a greater impact on their detection accuracy.

A bidirectional FPN (BI-FPN) is developed [51], which repeatedly applies top-down and bottom-up multiscale feature fusion to address the issues of backbone network information loss. Since the feature map formed after feature extraction is fully utilized, a multiscale feature map with stronger semantic and spatial information is obtained. However, overly complex detection networks do not always work well, especially in small ship detection. The more upsampling and downsampling there are, the more serious the feature information loss will be. Such a network structure will often cause negative effects.

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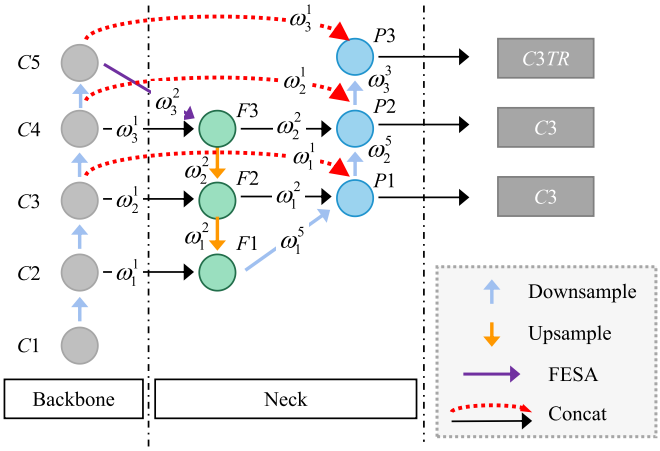


Fig. 6. Structure diagram of the proposed BiFF based on FESA.

Therefore, this article does not continue to perform upsampling after the $F1$ layer. The detailed structure diagram is shown in Fig. 6.

As shown in Fig. 6, the network structure uses the backbone network as the input features, and to enhance both the shallow and high-level semantic information of the predicted features, the feature maps of the backbone network are converted to the feature maps of the neck network for concatenation. In addition, different weights are assigned to the feature maps of each layer to determine their contribution. Feature maps for each layer can be described as follows:

$$F3 = \text{Concat}\left(\text{UpS}\left(\text{Conv}\left(\frac{\omega_3^2 \cdot (\text{FESA}(C5))}{\omega_3^2 + \xi}\right)\right), \omega_3^1 C4\right) \tag{16}$$

$$F2 = \text{Concat}\left(\text{UpS}\left(\text{Conv}\left(\frac{\omega_2^2 \cdot F3}{\omega_2^2 + \xi}\right)\right), \omega_2^1 C3\right) \tag{17}$$

$$F1 = \text{Concat}\left(\text{UpS}\left(\text{Conv}\left(\frac{\omega_1^2 \cdot F2}{\omega_1^2 + \xi}\right)\right), \omega_1^1 C2\right) \tag{18}$$

$$P1 = \text{DownS}\left(\text{Conv}\left(\frac{\omega_1^5 \cdot F1}{\omega_1^5 + \xi}\right), \omega_1^2 F\ 2, \omega_1^1 C3\right) \tag{19}$$

$$P2 = \text{DownS}\left(\text{Conv}\left(\frac{\omega_2^5 \cdot P1}{\omega_2^5 + \xi}\right), \omega_2^2 F\ 3, \omega_2^1 C4\right) \tag{20}$$

$$P3 = \text{DownS}\left(\text{Conv}\left(\frac{\omega_3^3 \cdot P2}{\omega_3^3 + \xi}\right), \omega_3^1 C5\right) \tag{21}$$

where $F1$, $F2$, and $F3$ represent intermediate features from level 1 to level 3 in the top-down path. $P1$, $P2$, and $P3$ represent the corresponding output features on the bottom-up path. $0 \leq w_i^j \leq 1$ $(i = 1, 2, 3, j = 1, 2, 3)$ represent different learnable weights that enable the network to learn the appropriate parameters in the iteration, and $\xi = 0.0001$ is used to avoid numerical instability. Conv represents the convolution operation. UpS and DownS represent the upsampling and downsampling operations, respectively. Concat represents the concatenate operation.

Three detection heads are designed in this article to detect the feature maps at different scales. Inspired by TPH-YOLOv5 [52], the transformer detection header captures global information, enriches contextual information, and improves the ability to capture different types of information. It can also explore feature representation potential using self-attentive mechanisms. This article argues that the feature maps become smaller after downsampling due to the anterior network structure. It is most severe on the third detection head, so C3TR is introduced into the third detection head to alleviate this problem. The overall structure of C3TR is shown in Fig. 7.

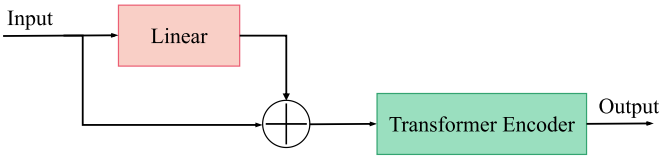


Fig. 7. C3TR structure diagram.

In summary, the new network structure uses weighted cross-scale connectivity to achieve higher level feature fusion first. Second, considering the complex detection network suitability on a small ship, upsampling is abandoned after $F1$. Third, the corresponding feature layers are introduced into the backbone network in the final detection head to enrich and supplement the feature information.

C. Coordinate Point Orientation IOU Loss Function

1) Existing Loss Function: The regression of the prediction box is a key factor in the process of target detection. Intersection over union (IOU) is a well-known evaluation metric that measures the degree of overlap between the prediction box and target box, and the loss function is defined as

$$L_{\text{IOU}} = 1 - \text{IoU}(A, B) \tag{22}$$

$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|} \tag{23}$$

where A represents the prediction box and B represents the target's real box.

However, as the IOU loss function only works well when the boundaries of the prediction box and the target's real box overlap, Distance IOU (DIOU) [53] is proposed to mitigate this issue by adding a penalty term that measures the distance between the centroid of the prediction box and the target box, and the loss function is defined as

$$L_{\text{DIOU}} = 1 - \text{IOU}(A, B) + \Re_{\text{DIOU}} \tag{24}$$

where $\Re_{\text{DIOU}}$ is the penalty term of the prediction box and the target box, which is calculated by

$$\Re_{\text{DIOU}} = \frac{\rho^2(d, d^{\text{gt}})}{c^2} \tag{25}$$

where d and d^{gt} are the centroids of A and B, respectively, $\rho(\cdot)$ is the Euclidean distance, and c is the diagonal length of the minimum enclosing box covering both boxes A and B.

It is worth noting that the ship size varies and the aspect ratio (width-to-height ratio) is also a major influencing factor. Complete IOU (CIOU) [53] is proposed by imposing the aspect ratio based on DIOU, and the loss function is

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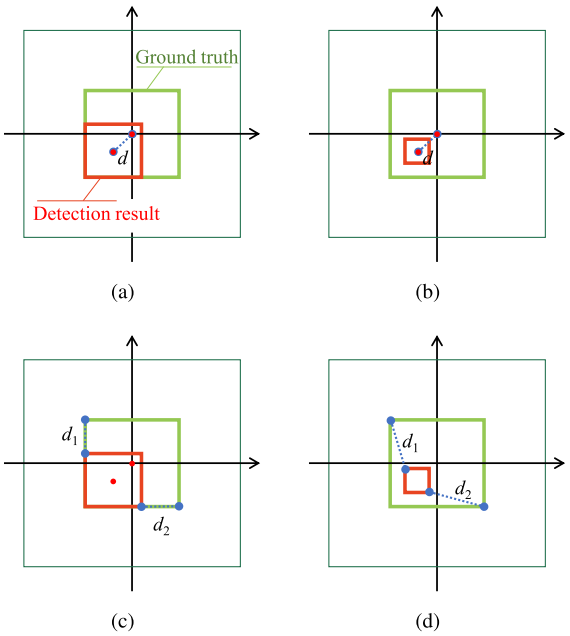


Fig. 8. Illustration of the limitation of existing loss functions and advantage of the proposed loss function. (a) and (b) Cases with the same distance between the centroid and aspect ratio. (c) and (d) Calculation of the above cases under CoIOU.

defined as

$$L_{\mathrm{CIOU}} = 1 - \mathrm{IOU}(A, B) + \Re_{\mathrm{CIOU}} \tag{26}$$

$$\Re_{\mathrm{CIOU}} = \Re_{\mathrm{DIOU}} + \alpha v = \frac{\rho^2(d, d^{\mathrm{gt}})}{c^2} + \alpha v \tag{27}$$

where α is the loss tradeoff parameter and v is used to measure the similarity of the aspect ratio

$$\alpha = \frac{v}{(1 - \mathrm{IOU}(A, B)) + v} \tag{28}$$

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w^{\mathrm{gt}}}{h^{\mathrm{gt}}} - \arctan\frac{w}{h}\right)^2. \tag{29}$$

The gradients of v are calculated as

$$\frac{\partial v}{\partial w} = \frac{8}{\pi^2}\left(\arctan\frac{w^{\mathrm{gt}}}{h^{\mathrm{gt}}} - \arctan\frac{w}{h}\right) * \frac{h}{w^2 + h^2} \tag{30}$$

$$\frac{\partial v}{\partial h} = -\frac{8}{\pi^2}\left(\arctan\frac{w^{\mathrm{gt}}}{h^{\mathrm{gt}}} - \arctan\frac{w}{h}\right) * \frac{w}{w^2 + h^2}. \tag{31}$$

For the CIOU loss function in (26), the aspect ratio describes the relative value, which is somewhat ambiguous for small target samples in specific cases, as shown in Fig. 8, where d denotes the centroid distance, the green box indicates the true value box, and the red box indicates the prediction box. It can be seen from the two cases shown in Fig. 8(a) and (b) that the aspect ratio and centroid remain the same, but the target size has changed. In other words, the IOU(A, B), the aspect ratio, and the centroid distance are the same. From the definition of the existing loss function, they cannot distinguish these two cases.

2) Proposed Loss Function: To address the problems of DIOU and CIOU, this article proposes a coordinate point orientation IOU loss function (CoIOU) to improve these defects, i.e., the corresponding Euclidean distances of the upper left and lower right endpoints of the prediction box and the target box are used as penalty terms. First, for the forward detection box, after the coordinates of the upper left endpoint and the lower right endpoint are determined, the centroid and aspect ratio of the box are then determined. Second, for the forward detection box, the centroid and aspect ratio of the box are determined after the coordinates of the upper left endpoint and the coordinates of the lower right endpoint are determined, which solves the problem of considering only the centroid but not the aspect ratio in DIOU. With the loss function proposed in this article, the width and height determined by the endpoint coordinates can compensate for the problem that the aspect ratio in CIOU does not apply to specific cases, and there is some ambiguity.

TABLE II
COMPARISON OF DIFFERENT RELATIONSHIP MEASUREMENT METRIC BETWEEN PREDICTION BOX AND GROUND-TRUTH BOX

Loss	Overlap Area	Centroid Distance	Aspect Ratio	Relative Size
L_{IOU}	✓			
L_{DIOU}	✓	✓		
L_{CIOU}	✓	✓	✓	
L_{CoIOU}	✓	✓	✓	✓

As shown in Fig. 8(c) and (d), the problem is solved when the Euclidean distances d_1 and d_2 between the corner point coordinates of the detection box are used for regression, which is equivalent to determining the relative size of the rectangular box through two points and, thus, indirectly determining the centroid and aspect ratio. Table II shows the comparison of different relationship measurement metrics between the prediction box and the ground-truth box.

In the loss function proposed in this article, the penalty term becomes expressed as follows:

$$\Re_{\mathrm{CoIOU}} = \frac{\rho^2\left(d_1, d_1^{\mathrm{gt}}\right)}{2c^2} + \frac{\rho^2\left(d_2, d_2^{\mathrm{gt}}\right)}{2c^2} \tag{32}$$

where d_1 denotes the coordinates of the top-left corner point of the prediction box, d_1^{gt} denotes the coordinates of the top-left corner point of the target ground-truth box. d_2 denotes the coordinates of the bottom-right corner point of the prediction box, and d_2^{gt} denotes the coordinates of the bottom-right corner point of the target ground-truth box. c is the diagonal length of the minimum enclosing box covering both boxes. Then, the loss function proposed in this article can be converted to

$$L_{\mathrm{CoIOU}} = 1 - \mathrm{IOU}(A, B) + \frac{\rho^2\left(d_1, d_1^{\mathrm{gt}}\right)}{2c^2} + \frac{\rho^2\left(d_2, d_2^{\mathrm{gt}}\right)}{2c^2}. \tag{33}$$

Moreover, due to the SAR imaging mechanism, the influence of the presence around the ship hull may lead to blurring of the target centroid, which leads to poor fitting of the target prediction box and the real box. In this article, the proposed loss function is divided into two parts: one is the loss function that retains the IOU, and the other is the newly proposed distance loss. The new distance loss is considered to be able to

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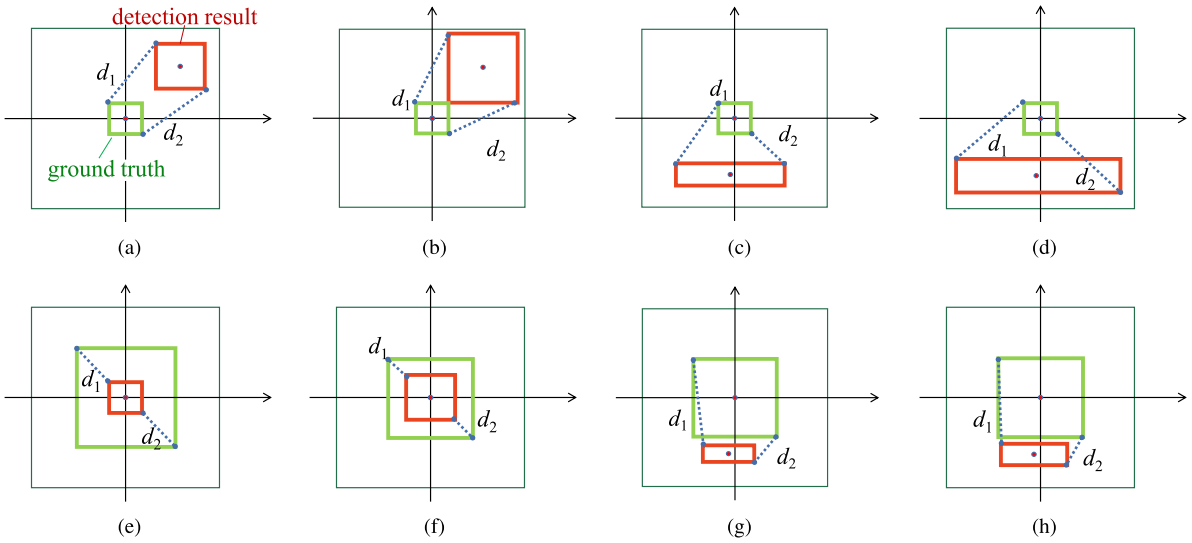


Fig. 9. Adaptability of the proposed CoIOU loss in different cases. (a)–(d) Ground truth is smaller than the detection result. (e)–(h) Ground truth is larger than the detection result.

use the IOU for comprehensive determination while relying on the distance of two coordinate points for regression to assist in determining the accuracy of the detection box. In this way, the detection rate, the false alarm rate, and the fit of the target are improved while considering the sidelobes and the fit. In this article, most of the cases are shown in Fig. 9.

For the example in the Large-Scale SAR Ship Detection Dataset (LS-SSDD), this article applies the theory to the small target, and in the regression of the prediction box, which singles out the small target with obvious sidelobe for analysis, finds that, with the loss function proposed in this article, the prediction box of the small ship can be more fully considered. This article shows the regressions that may occur when the detection frame corresponds to a small ship, as shown in Fig. 10. From the theoretical prediction, it can be seen that the detection box can be regressed to fit the target using the CoIOU loss function for small ships of different orientations. By shortening the Euclidean distance between the coordinates of the upper left corner point and the coordinates of the lower right corner point in different cases, the influence of the ship hull sidelobe is indirectly weakened, and the prediction box is closer to the target box under one iteration.

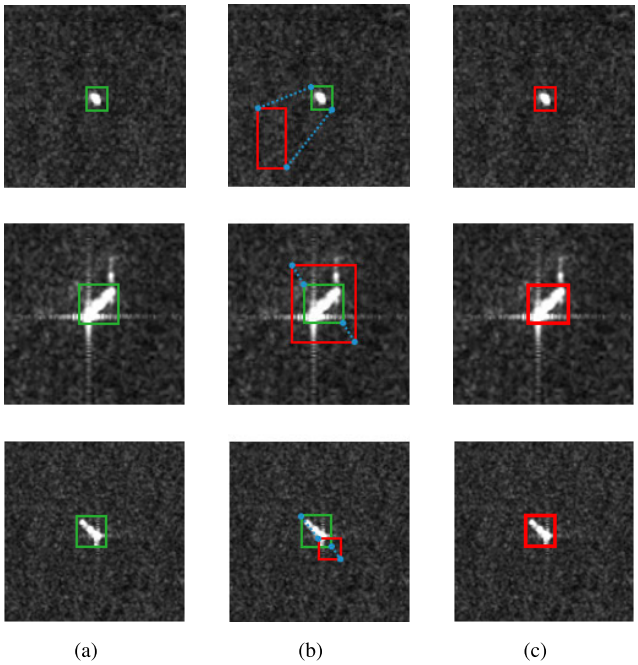


Fig. 10. Actual possible scenarios of detection boxes for small targets in the LS-SSDD dataset. (a) Real box of the target. (b) Possible scenarios in different cases. (c) Final detection results that will be obtained.

III. Experiment and Discussion

To fully validate the method proposed in this article, the detection performance of the proposed article is validated using three publicly available ship datasets: the AIR-SARShip, the SAR Ship Detection Dataset (SSDD), and LS-SSDD, respectively. This section begins with a brief introduction to the datasets and experimental data arrangement of this article, followed by the performance metrics of the evaluated models. The superiority of the proposed method is demonstrated by comparison experiments and ablation experiments.

A. Experimental Data

The three datasets, including AIR-SARShip, SSDD, and LS-SSDD, contain representative SAR ship data, which include data from large ships, small ships, near-shore ships, and far-shore ships. Their information is shown in Table III. In this study, small ships are defined according to the reference information provided in the LS-SSDDv1.0 dataset, where M is defined as the number of pixels of an image, and then, the ship with a label box area less than $M \times 0.37\%$ is defined as a small ship target. Hence, a ship calling for a label box area smaller than $800 \times 800 \times 0.37\% = 2368$ is defined as a small ship target.

1) AIR-SARShip [15] contains 31 large-scale scenes. The image size is an overwhelmingly 3000×3000 pixels, and the training set is reduced to 512×512 pixels in this

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TABLE III
DETAILS OF DATASET

Parameter	AIR-SARShip	SSDD	LS-SSDD
Satellite	Gaofen-3	RADARSAT-2, TerraSAR-X, Sentinel-1	Sentinel-1
Resolution	1m, 3m	1m∼15m	5m×20m
Image Size (pixel)	3000×3000	416×323, 501×355, etc.	24,000×16,000
Polarization	Single	HH, HV, VV, VH	VV, VH
Swath (km)	9	10	250
Image number	31	1160	15

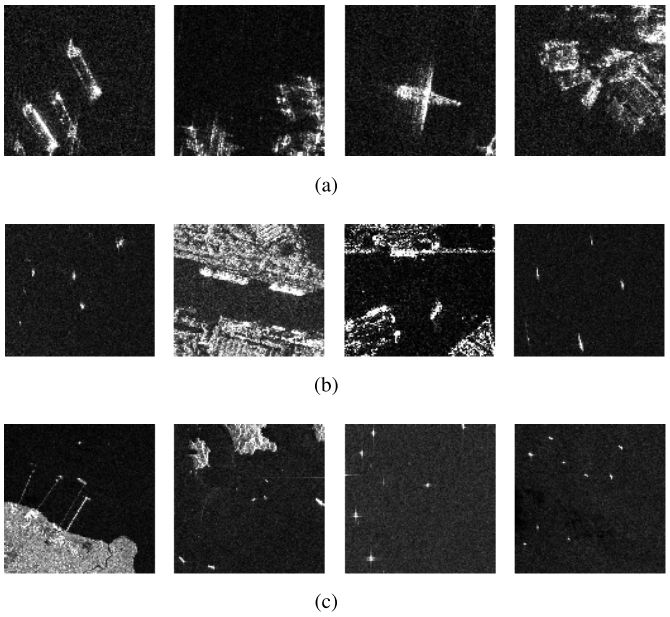


Fig. 11. Sample images in the three experiment datasets. (a) AIR-SARship. (b) SSDD. (c) LS-SSDD.

article, as shown in Fig. 11(a). A total of 1080 images are used as a training set, and 120 images are used as a testing set in this article. The image resolution ranges from 1 to 3 m. The imaging modes include a spotlight and strip map. The polarization mode is single polarization. There are 512 ships in total in AIR-SARShip. Among them, there are 74 small ships, 406 medium ships, and 32 large ships.

2) The SSDD [16] dataset images are primarily from RADARSAT-2, TerraSAR-X, and Sentinel-1 sensors, with four polarization modes of HH, HV, VV, and VH, and the dataset contains 1160 SAR images of various image sizes, such as 416 × 323 and 501 × 355, as shown in Fig. 11(b). A total of 928 images are used as a training set, and 232 images are used as a testing set in this article. The resolution ranges from 1 to 15 m. There are 2551 ships in total in SSDD. Among them, there are 1463 small ships, 989 medium ships, and 99 large ships.

3) The LS-SSDD [17] dataset images are taken by Sentinel-1, which consists of 15 images with a size of 24 000 × 16 000 pixels. The large-scale image is cut into 9000 subimages with a size of 800 × 800 pixels in this article, as shown in Fig. 11(c). A total of 6000 images are used as a training set, and 3000 images are used as a testing set in this article. The image resolution is 5 × 20 m. There are 6015 ships in total in LS-SSDD-v1.0. Among them, there are 6003 small ships and 12 medium ships.

B. Hyperparameters and Environment Settings

In this article, the parameters are set empirically. The network is trained using the stochastic gradient descent (SGD) approach, with the learning rate set at 0.001. Momentum is set to 0.937, and the weight decay is set to 5e-4. The anchors are set to [1:1, 1:2, 2:1] for three aspect ratios, the anchor size is set to 15, the epoch is set to 100, and the learning rate is reduced to 0.0001 at the 50th epoch to facilitate the search for the optimal model. In addition, the software environment is Pytorch-1.7.1, CUDA 10.1, and cuDNN 7.6.5, and the hardware environment is given as follows: the CPU is Intel Core i7-10700K, and the GPU is GeForce RTX 2070.

C. Evaluation Metrics

In this article, precision, recall, and average precision (AP) are used to evaluate the performance of the detection model. The precision is defined as the proportion of correctly detected ships among all detected ships

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{34}$$

where TP represents true positives, the total number of ships correctly detected, and FP represents false positives, the total number of ships incorrectly detected, indicating false alarms. Recall indicates the fraction of detected ships relative to all ground truth, denoted as

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{35}$$

where FN represents false negatives, also known as missed detection. It can be seen that recall and precision are interrelated to influence each other.

AP is also determined by recall and precision, and is defined as

$$\text{AP} = \int_0^1 P(R) dR. \tag{36}$$

$P(R)$ is expressed as the curve of precision and recall, which can be interpreted as the area contained by the $P(R)$ curve and the coordinate axis. The AP score in this article is the AP under the IOU score of 0.5 of the detection box.

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TABLE IV
COMPARISON WITH OTHER METHODS ON LS-SSDD DATASET

Method	Precision(%)	Recall(%)	AP(%)
Faster R-CNN [18]	73.74	77.71	74.80
YOLOv3 [54]	**87.64**	26.24	24.63
Cascade R-CNN [19]	84.09	72.67	70.88
DCN [55]	74.13	76.87	73.84
Libra R-CNN [56]	73.55	76.70	73.68
SAR-Net [38]	58.37	**81.96**	76.20
LTD-RBD [57]	/	81.00	71.20
L-YOLO [58]	84.78	63.96	73.01
MSIF [37]	86.58	78.17	75.57
Ours	86.30	80.00	**80.30**

TABLE V
COMPARISON WITH OTHER METHODS ON SSDD DATASET

Method	Precision(%)	Recall(%)	AP(%)
Faster R-CNN [18]	85.37	90.07	89.15
DAPN [59]	91.36	85.54	90.56
Quad-FPN [32]	95.77	89.52	95.29
Cascade R-CNN [19]	90.81	94.10	90.50
Double-Head R-CNN [20]	91.91	86.96	91.10
HR-SDNet [60]	90.99	**96.49**	90.82
PANET [61]	91.91	86.81	91.15
Grid R-CNN [62]	89.71	87.77	88.92
SER Faster R-CNN [63]	92.28	86.11	91.52
Ours	**96.10**	94.00	**97.80**

TABLE VI
COMPARISON WITH OTHER METHODS ON AIR-SARSHIP DATASET

Method	Precision(%)	Recall(%)	AP(%)
NNAM [64]	66.10	79.40	70.80
DCMSNN [65]	67.20	78.50	71.50
SSD [20]	66.70	71.50	69.80
YOLOv3 [54]	67.30	70.60	67.40
EFGRNet [65]	66.60	72.50	70.30
Faster R-CNN [18]	64.80	78.30	73.70
Cascade R-CNN [19]	67.80	79.60	73.60
Libra R-CNN [56]	64.80	78.30	72.50
CenterNet+ [66]	68.00	**83.60**	73.90
Ours	**70.00**	79.00	**79.00**

D. Overall Detection Performance Comparison

The proposed method in this article is compared with some state-of-the-art methods in deep learning. Tables IV–VI show the final ship detection metrics for each of the three datasets. The method presented in this article has the highest AP, reaching 80.30% on LS-SSDD datasets. On the SSDD datasets, this article's method has the highest precision and AP, reaching 96.10% and 97.80%, respectively. On the AIR-SARShip datasets, this article's method has the highest precision and AP, reaching 76.50% and 79.00%, respectively.

The proposed method makes use of three contributions: FESA, BiFF based on FESA, and CoIOU. FESA reduces the effects of strong scattering points outside of the ship body and enhances the ship body information. The new network structure, BIFF based on FESA, compensates for the information lost during FESA. CoIOU, a novel loss function, accurately describes various overlapping box situations. The efficacies of FESA and CoIOU are analyzed by experiments in this section. The efficacy of BiFF based on FESA will be shown during the ablation experiment in Sections III-D1 and III-D2 since the proposed network structure is constructed based on FESA.

TABLE VII
DETECTION METRICS OF DIFFERENT FEATURE EXTRACTION METHODS ON SSDD DATASET

Method	Precision(%)	Recall(%)	AP(%)
Baseline	94.00	86.60	93.70
CBAM [48]	93.30	**91.40**	96.10
FESA (Ours)	**95.90**	90.50	**96.50**

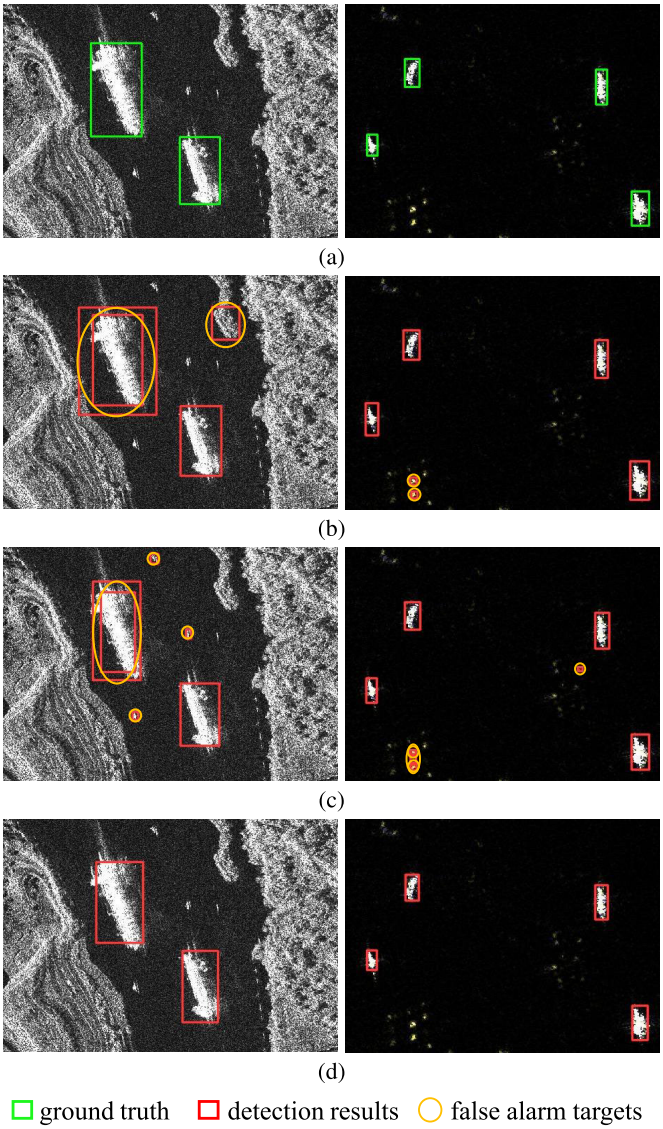
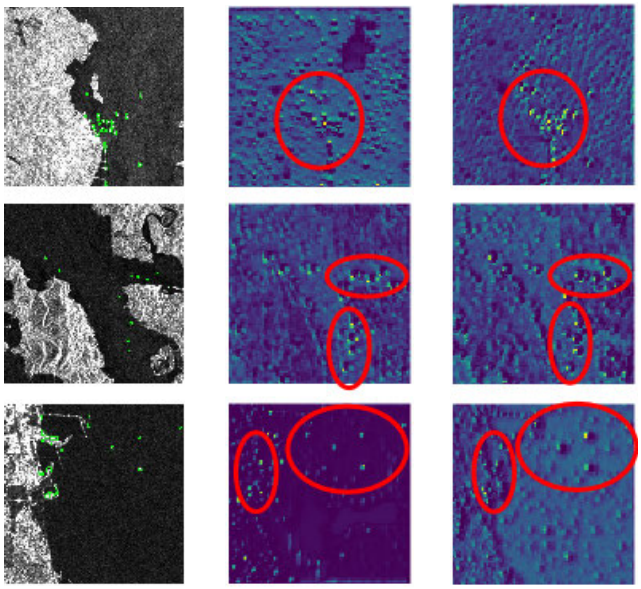


Fig. 12. Testing the performance of the FESA. Green boxes indicate ground truth, red boxes indicate detection results, and yellow circled boxes indicate false alarm targets. (a) Ground truth. (b) Baseline. (c) CBAM [48]. (d) FESA proposed in this article.

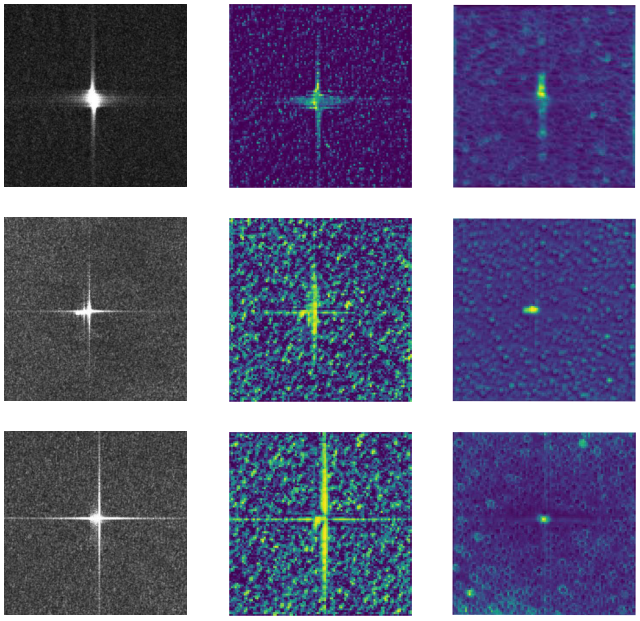
1) Efficacy Analysis of FESA: According to (9) and (10), FESA first uses maximum pooling and average pooling on the feature map in two directions. Average pooling can take

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(a) (b) (c)

Fig. 13. Feature maps after FESA. (a) Real ship labels on the LS-SSDD dataset. (b) Output of feature maps under YOLOv5l. (c) Output of feature maps under the FESA.



(a) (b) (c)

Fig. 14. Small ship slice's feature maps. (a) Original image of the small ship slice. (b) Feature map without FESA. (c) Feature map with FESA.

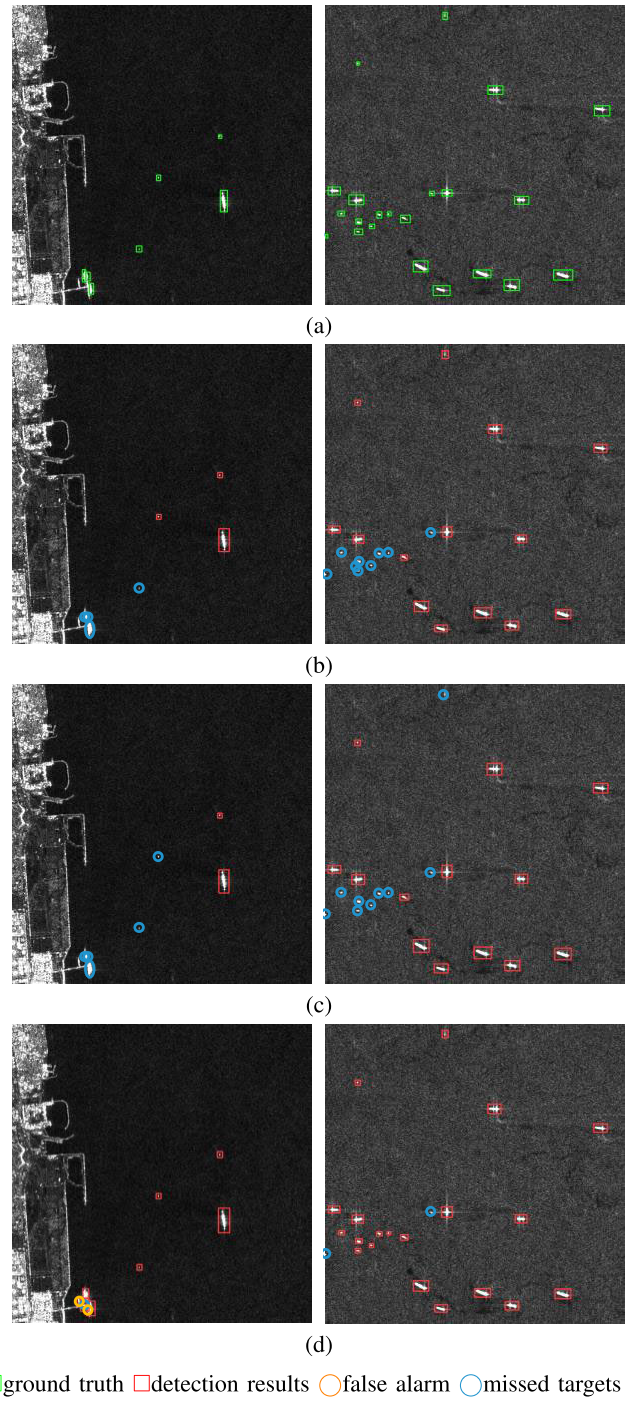


Fig. 15. Testing the performance of CoIOU. Green boxes indicate ground truth, red boxes indicate detection results, blue circled boxes indicate missed targets, and yellow circled boxes represent false alarm targets. (a) Ground truth. (b) DIOU. (c) CIOU. (d) CoIOU proposed in this article.

the mean value from a region's pixels and weaken the higher value in the region, while maximum pooling can extract the maximum value from a region's pixels. Finally, (15) is able to integrate the maximum pooled features and the average pooled features to obtain the feature map through FESA. The following experiments are carried out to confirm the efficacy of the proposed FESA. The baseline detection model is YOLOv5l, and the FESA is added after the backbone and tested on the SSDD dataset. The detection metrics are shown in Table VII.

In Table VII, CBAM is based on the baseline, and the attention mechanism module of CBAM is added after the backbone. After that, the CBAM module is replaced by the feature extraction method proposed in this article. It can be seen that AP is improved by 2.4% over baseline after introducing the imaging mechanism-guided feature extraction method. Compared with CBAM, AP also improves by 0.4%. Both of them have improved to a great extent in terms of accuracy. The results validate that the proposed FESA in this

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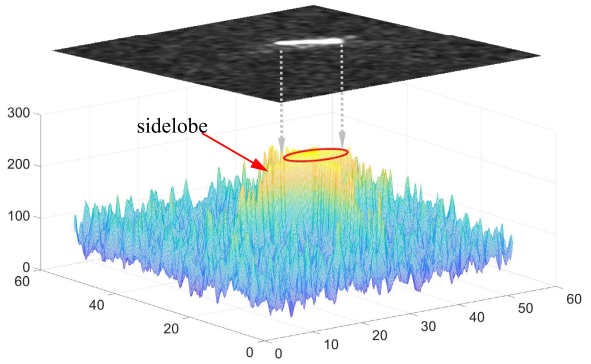
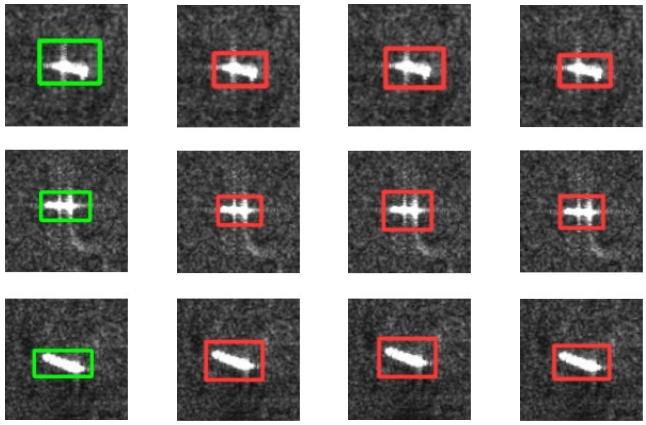


Fig. 16. 3-D display of the sidelobe of a small ship.



(a) (b) (c) (d)

Fig. 17. Selected ship targets under the magnified observation of the fit of the target box under different loss functions. (a) Ground truth. (b) DIOU. (c) CIOU. (d) CoIOU proposed in this article.

article improves the ship detection performance and increases the target localization accuracy.

To visually illustrate the effect of imaging-mechanism feature extraction, some detection results are shown in Fig. 12. Fig. 12(a) shows the target real box. There is no target in the lower left corner of the image on the left-hand side of Fig. 12(a), but there are clear false alarms (marked with yellow circles) in Fig. 12(b) and (c). On the right-hand side of Fig. 12(a), there are no targets except for two large ships, but there are obvious false alarms (marked by yellow circles) in Fig. 12(b) and (c). In Fig. 12(d), for both images, the real labels in the original image are detected completely and correctly with the FESA without any missed detections or false alarms.

In addition, to verify the effectiveness and applicability of the proposed method, some images are selected on the LS-SSDD dataset for the refined representation of the feature maps. As shown in Fig. 13, after using the method proposed in this article, the corresponding ship hull information of the feature map has been improved significantly in the red circle. To represent the effect of FESA more accurately, we used small ship slices from LS-SSDD, conducted the experiment separately, and visualized the intermediate feature maps to demonstrate the effect of FESA. As shown in Fig. 14, it can be clearly seen that the sidelobe interference is effectively

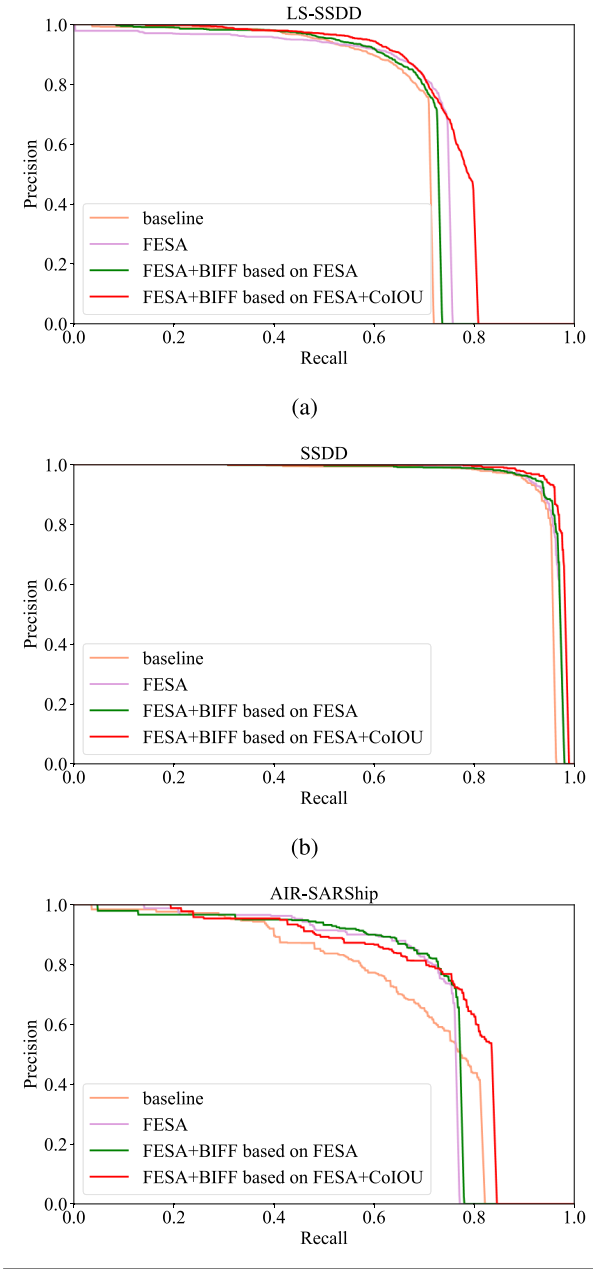


Fig. 18. PR curves of ablation experiments on different datasets. (a) LS-SSDD. (b) SSDD. (c) AIR-SARShip.

improved in (c), which fully demonstrates its much improved feature recognition capability for small ship targets and in complex scenes. Otherwise, sidelobe interference that is too strong may make the detection frame too large or form missed detections and false alarms.

2) Efficacy Analysis of CoIOU: As we argued in Section II, the novel loss function is proposed based on the dual Euclidean distances between the corner coordinates of the prediction box and the real box since the conventional loss function based on centroid distance and aspect ratio may result in the same loss function value for different prediction box sizes. The corresponding Euclidean distances of the upper left and lower right endpoints of the prediction box and the target box are

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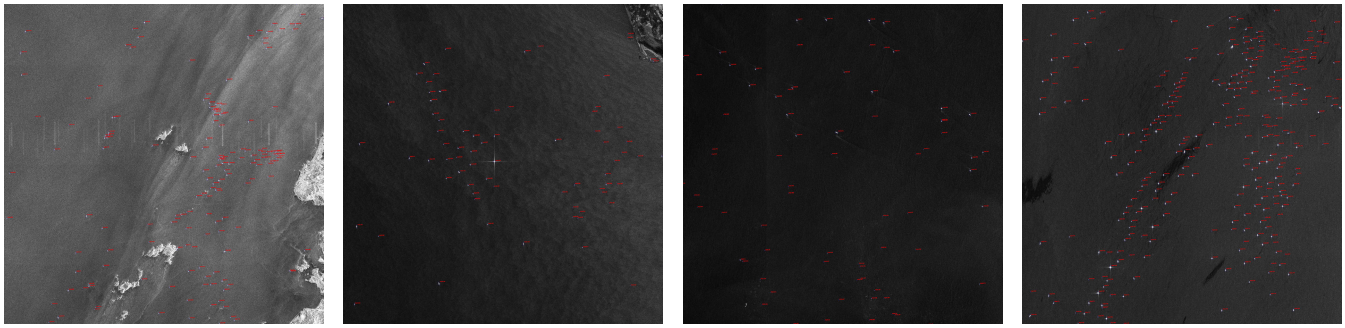


Fig. 19. Large-scale areas (24 000 * 16 000) detection results.

TABLE VIII
DETECTION METRICS OF DIFFERENT LOSS FUNCTIONS ON LS-SSDD DATASET

Method	Precision(%)	Recall(%)	AP(%)	AP_{95}
DIOU	82.70	64.50	75.30	29.20
CIOU	**84.60**	65.70	78.00	31.50
CoIOU(Ours)	83.20	**69.90**	**78.50**	**32.70**

TABLE IX
ABLATION EXPERIMENTS ON THREE DATASETS

FESA	BiFF based on FESA	CoIOU	AP(%)		
			AIR-SARShip	SSDD	LS-SSDD
-	-	-	70.30	93.70	78.00
•	-	-	78.50	96.50	78.50
•	•	-	78.80	96.70	79.00
•	•	•	79.00	97.80	80.30

used as penalty terms, as in (33). Especially in the detection of small ships, this can solve the problem of considering only the centroid but not the aspect ratio in DIOU. With the loss function proposed in this article, the width and height determined by the endpoint coordinates can compensate for the problem that the aspect ratio in CIOU does not apply to specific cases, and there is some ambiguity in order to improve the performance of small ship detection. On the LS-SSDD dataset, the proposed loss function is assessed, and its performance is compared to that of DIOU and CIOU.

Fig. 15 shows some visual detection results by DIOU, CIOU, and CoIOU. When using CIOU and DIOU, a few small ships in the lower left of the image are not detected, whereas, with CoIOU, this missed detection is considerably improved.

The quantitative detection metrics are shown in Table VIII. The regression loss function proposed in this article should also significantly increase the degree of fit between the detection box and ground-truth box, which can be measured by AP_{95}. AP_{95} is the average score when the detection box overlaps with the ground-truth box by more than 95%, i.e., IOU = 0.95, so the criterion is stricter. The comparison in Table VIII reveals that the AP score of CoIOU is the highest, while AP_{95} is likewise much higher than those of CIOU and DIOU.

The small ship targets in the image are selected for a 3-D display, as shown in Fig. 16. The quantization is reduced so that the impact from the ship body and the sidelobes can be clearly seen. This enables a clearer view of the effect of the sidelobe on the small ship targets. In addition, some typical targets are selected and shown in Fig. 17. The effects of the three loss functions after the detection box is enlarged are compared. It can be seen that, under the CoIOU loss function, the detection box fits the ship target even if the original target has certain sidelobe effects. The degree of the detection box fit of the ship target is also improved by the CoIOU loss function. This also proves the effectiveness of CoIOU for small ship detection.

E. Ablation Experiment

The ablation experiments are performed on the three datasets sequentially by the three contribution points in this article, while the model structure of this article is based on the FESA, as shown in Table IX. The final results obtained by superimposing the three solutions are 80.3% on the LS-SSDD dataset, 97.80% on the SSDD dataset, and 79.00% on the AIR-SARShip dataset. The comparison of the accuracy–recall curves for the three datasets used in the ablation experiments is shown in Fig. 18(a)–(c). The curves make it clear that, thanks to the example provided by the ablation experiments, superimposing the three approaches also results in an increase in the curve envelope area, which validates the module's efficacy.

F. Large-Scale Areas' Robust Experiment

To verify the detection performance of this method in large-scale areas, we selected five large-scene images from Sentinel-1. The resolution of all five images was 24 000 * 16 000, and the final AP value was 75.3%. Large-scale areas showed a slight but acceptable decrease in performance compared to slicing detection. This also fully illustrates the robustness of our experimental model. We show some detection results on large-scale areas in Fig. 19.

IV. CONCLUSION

In this article, an SAR imaging mechanism-guided small ship detection network with sidelobe awareness and dual-coordinate regression is proposed. The introduction of a new imaging feature-guided feature extraction for the ship target effectively extracts the local features of the ship, which

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can weaken the influence of the sidelobes on the ship body to a certain extent and better extract the ship body information, making the model more applicable to SAR image datasets. In addition, the proposed model structure based on FESA can use contextual information to better extract image semantic information. Finally, the loss function with full consideration is constructed to attenuate the false alarm rate and missed detection rate, and improve the fitting effect of the detection box. The experimental results show that the method proposed in this article can improve the detection performance of small ship targets in SAR images and reduce the effects of false alarms and missed detection of ship targets. In the future, the testing speed of the method may be further studied and improved to obtain better detection results.

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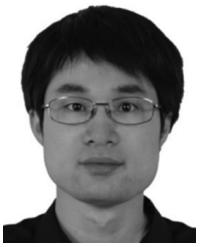
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